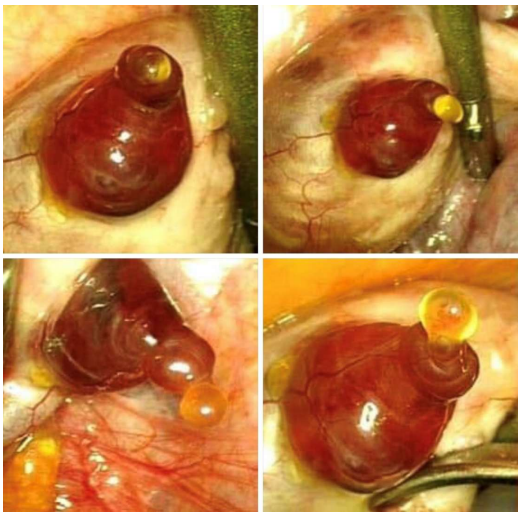
Histology of the Ovary and Organs of the Ovum Pathway



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Image: laparoscopic view of an ovary during ovulation, with the stigma visible and extruding follicular fluid.

LEARNING OBJECTIVES

By exam time, you should be able to do the following:

- Discuss the structural and functional anatomy of the ovary and organs involved in ovum release or implantation, including how they change over different life stages and the hormonal regulation of these changes.
- Describe and identify the cells and tissues of the ovary and organs involved in ovum release or implantation by histology.
- Discuss the process of oogenesis from primordial follicle through mature oocyte, including the associated morphologic changes.
- Compare and contrast how the ovary and organs involved with ovum release or implantation function as, and are regulated by, endocrine glands; and identify the impact of hormonal change via microscopy.

Starting in academic year 2025–2026, this learning activity was changed from a histology lab to a lecture. This was done in response to student requests for more structure regarding this content. However, the resources from the session as a histology lab still exist!

The materials are included on Canvas as optional, supplementary material that give you an opportunity to practice the content within this session.

The Ovary

A note on terminology, before diving into the histology of the ovary: you may see differences in how various resources (and our own lecturers!) discuss the stages of follicular development. This probably represents a difference between sources for the material. For this anatomy lecture, I am using the assigned textbook (Pawlina & Ross, *Histology: A Text and Atlas*, chapter 23) as my source. In this reference, it provides a disclaimer:

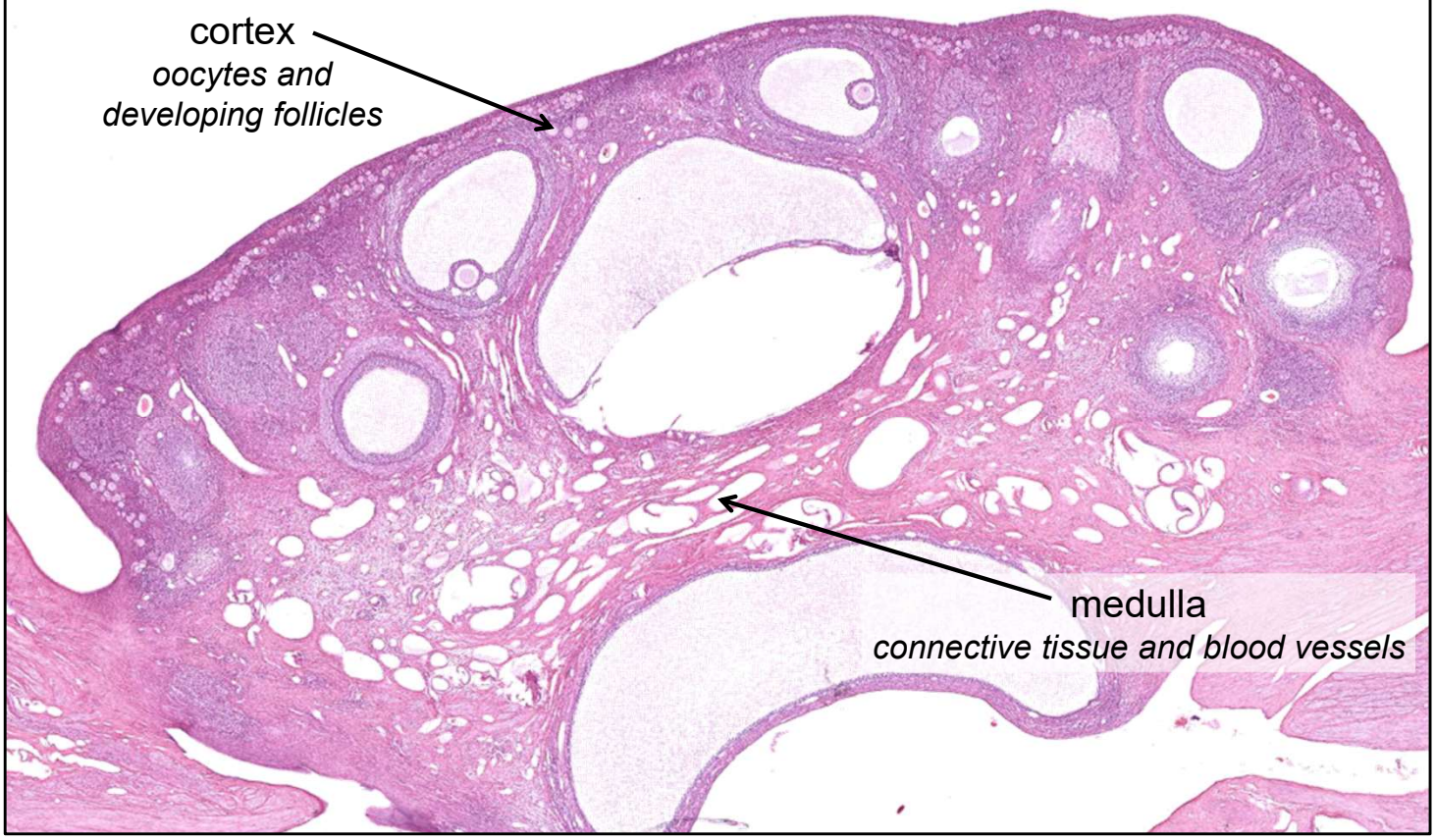
"*Histologically, three basic types of* ***ovarian follicles*** *can be identified on the basis of developmental state:*
Primordial follicles
Growing follicles*, which are further subcategorized as primary and secondary (or antral) follicles*
Mature follicle *or* ***Graafian follicles***

Some histologists and clinicians identify additional stages in the continuum of follicular development."

The Pawlina text defines a primary follicle as having simple cuboidal cells (early) which subsequently stratify (late); a secondary follicle as containing a fluid-filled antrum. However, other sources may use slightly different criteria for determining these different levels of maturation.

For our purposes, we're earmarking this as a possible point of confusion for students and will make sure to avoid assessment items that might lead to having more than one correct answer based on the diversity of sources.

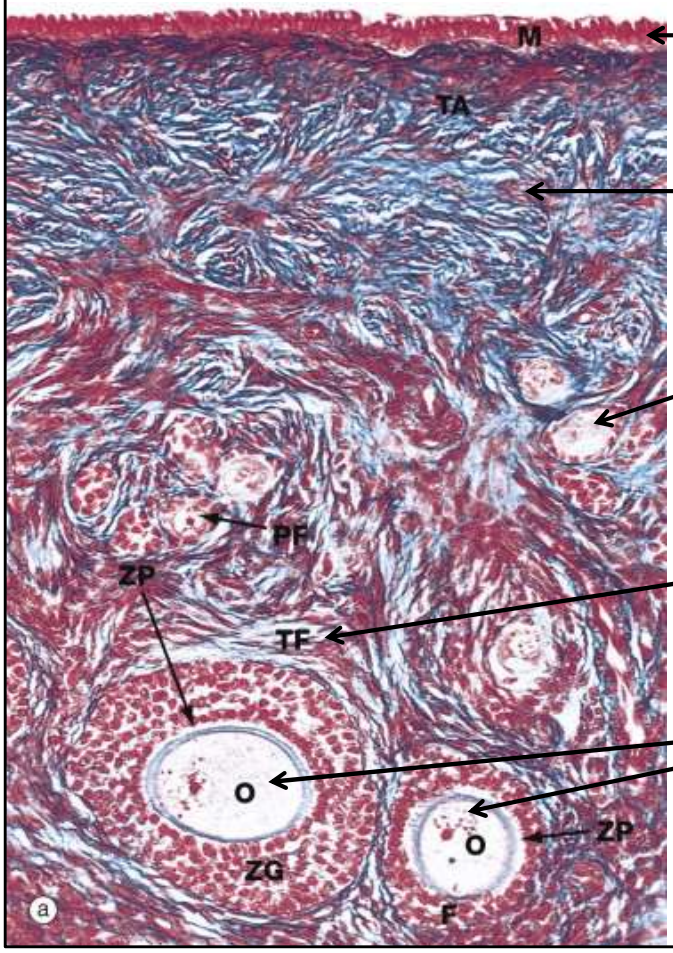
MICROSCOPIC STRUCTURE OF THE OVARY



This histology lab will trace the structures of the ovum pathway along the course an egg would take through the body. The **ovary** is the **gonad** that produces **oocytes** (**ova** or **egg cells**). It can be subdivided, like many organs, into a cortex and a medulla. The **ovarian cortex** is the primary parenchymal region of the organ and houses the developing **follicles** and the oocytes they contain. It accounts for most of the gonad by volume. The cortex is in contact with the overlying peritoneum, which—in this region—gets a special name. The peritoneum covering the ovary, fallopian tubes (oviducts), and uterus is called the **broad ligament** and can be subdivided into constitutive parts. The portion of the broad ligament covering and suspending the ovary is termed the **mesovarium**.

The **ovarian medulla** is the deeper portion of the organ and predominantly houses connective tissue and blood vessels that support the overlying cortex. It is a smaller contributor to the overall ovarian volume.

OVARIAN CORTEX: STRUCTURE AND PRIMARY FOLLICLES



mesothelium (mesovarium)
single layer of cells on surface

tunica albuginea
encapsulates ovary

primordial follicle
primary oocyte surrounded by single layer of follicular cells

theca folliculi
stromal cells

primary follicles
once stimulated, oocyte enlarges as follicular cells become cuboidal and proliferate; zona pellucida forms between oocyte and follicular (now granulosa) cells

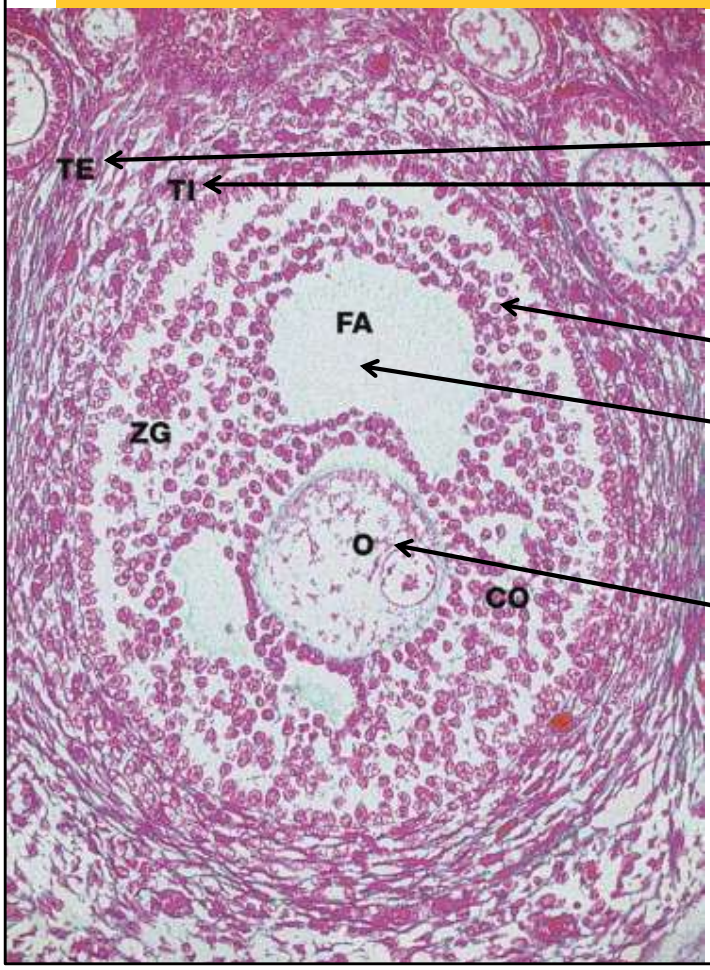
As mentioned on the previous slide, the ovarian cortex is covered with a layer of peritoneum, more specifically termed the **mesovarium**. This is a single layer of cells on the surface, which generally take on a squamous to cuboidal morphology. Historically, this layer has also been called the **germinal epithelium** because it was believed that the oocytes derived from this layer before entering the substance of the ovary. We now know that isn't the case but the term lives on. Just deep to the mesothelium is a layer of dense irregular connective tissue called the **tunica albuginea** ("white shirt" or "white membrane" in Latin) that provides a structural capsule to the organ.

Within the parenchyma of the organ can be found follicles and the oocytes they contain undergoing development. We'll trace the different stages in that development in this slide and the remainder of this section. The earliest form in development is the **primordial follicle**, which contains a **primary oocyte** surrounded by a single layer of squamous **follicular cells**. This primary oocyte is arrested in **prophase of meiosis I** and will demonstrate a large nucleus with a dark nucleolus, as well as abundant organelles (Golgi, endoplasmic reticulum, and so on), giving it its large size. Note that the primordial follicles formed during embryogenesis and represent the pool of available oocytes; it is believed that there is no additional stem cell compartment within the ovary. An individual is born with all the follicles in this stage of development, and they remain arrested as such until **menarche** (an individual's first **menstrual cycle** or **period**).

At that point, once every approximately 28 days, a small group of primordial follicles (~20–30) are recruited to continue maturation and become **primary follicles**. The first indication of this advancement in development is that the follicular cells change their morphology from being squamous to cuboidal. This time point may be called a **unilaminar primary follicle**. Thereafter, the follicular cells become proliferate under the stimulation of **follicle-stimulating hormone** (**FSH**) and become multilayered, representing a **multilaminar primary follicle**. At this point, we change the name of follicular cells to be called **granulosa cells**. In addition, a thick layer of glycoproteins forms between the oocyte and the granulosa cells, called the **zona pellucida** ("transparent zone" in Latin). Granulosa cells have extensive gap junctions between themselves, allowing communication, and connect to the oocyte through processes that pierce the zona pellucida.

Present in the background, as stromal cells around the follicles, is the **theca folliculi**. As we will see on the next slide, the theca will become organized around the developing follicles as they continue development and serve endocrinological roles.

OVARIAN CORTEX: SECONDARY FOLLICLE



theca folliculi
forms two layers:
***theca externa**: capsule-like conn. tissue layer*
***theca interna**: produces androstenedione*
(converted to testosterone
and estradiol by follicular cells)

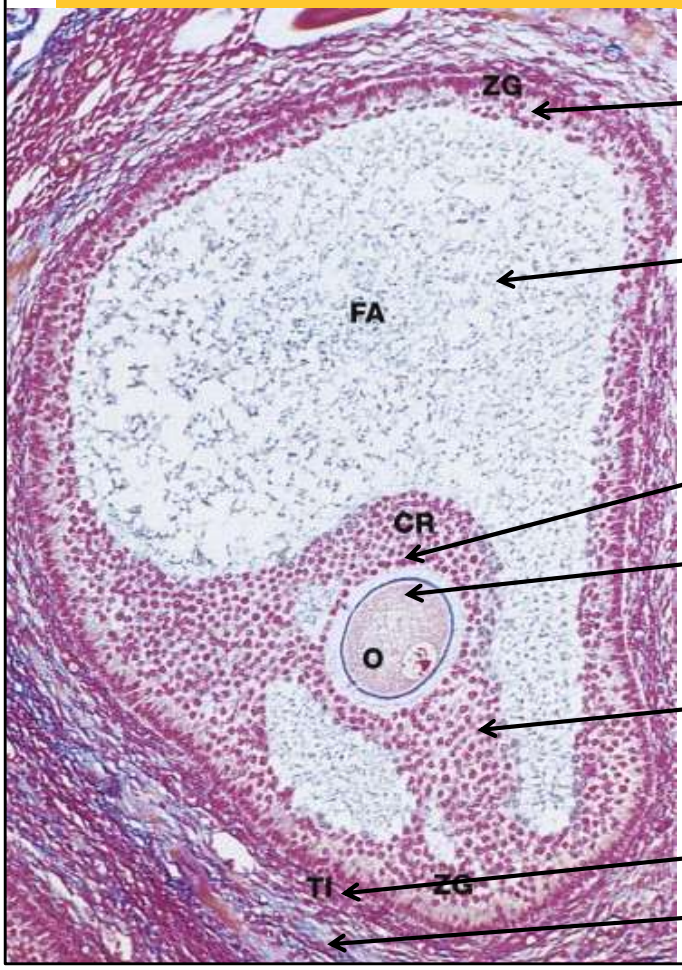
granulosa cells
proliferate and form fluid-filled
follicular antrum

oocyte

As the primary follicles continue to develop, they become **secondary follicles**. This transition is marked by the formation of a fluid-filled **follicular antrum**, which shows up as many small white spaces originally that eventually coalesce to form a large cavity between the granulosa cells, now roughly 6–12 layers thick. The **follicular fluid** (**liquor folliculi**) contains abundant hyaluronic acid. This time point also represents when the oocytes gradually begin to re-enter meiosis, overcoming their arrest at prophase I. The granulosa cells themselves continue to proliferate, forming a few specific regional groups that will be described further on the next slide.

Here, we will also comment on the formation of two layers of the **theca folliculi**, which surrounds the follicle and is separated from the granulosa cells by a basement membrane. The **theca interna** surrounds the granulosa cells and is separated from them by a distinct basal lamina. Under **luteinizing hormone** (**LH**) stimulation, the theca interna cells produce **androstenedione**. The granulosa cells take this androstenedione and eventually convert it into **estrogens**; this requires the enzyme **aromatase**, which the granulosa cells are stimulated to produce by **follicle-stimulating hormone** (**FSH**). Around the theca interna is the **theca externa**, which is composed of spindle-shaped cells resembling fibroblasts and some smooth muscle. Later in the development of the oocyte, these cells will be responsible for mechanically extruding the mature oocyte during ovulation.

OVARIAN CORTEX: GRAAFIAN (MATURE) FOLLICLE



zona granulosa
granulosa cells surrounding follicular antrum

follicular antrum

corona radiata
granulosa cells surrounding oocyte

oocyte
completes meiosis I right before ovulation, then arrests in metaphase II until fertilization

cumulus oophorus
granulosa cells anchoring corona radiata to zona granulosa

theca interna

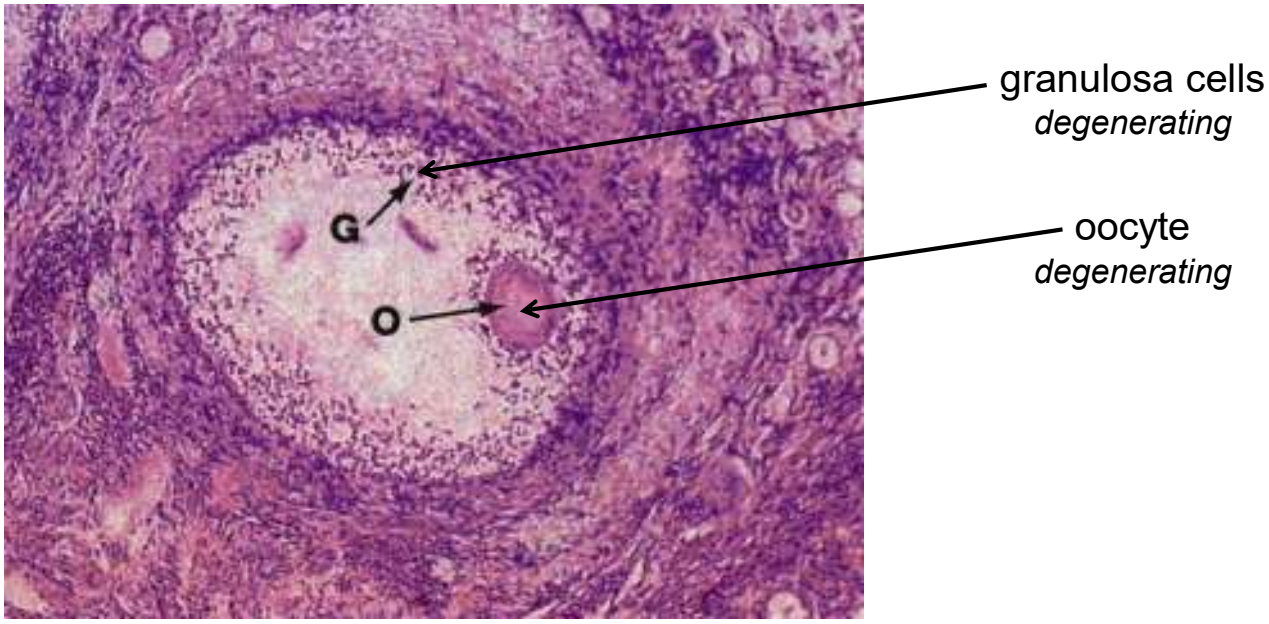
theca externa

As the **follicular antrum** expands, the follicle becomes a **mature** or **Graafian** follicle (once it is >10 mm in diameter). The oocyte has also continued through meiosis I in the interim and will progress into meiosis II, arresting again at **metaphase of meiosis II**. In the process of the first meiotic division, it will leave behind a **polar body** which contains a haploid complement of chromosomes, but minimal cytoplasm. (The oocyte takes as much of the cytoplasm, intracellular organelles, and mRNA for itself as possible.) Let's examine the granulosa cells, now that we can appreciate their three groups a bit better:

- The **corona radiata** corresponds to the layers of cells immediately surrounding the oocyte.
- The **zona granulosa** lines the external aspect of the follicle, including of the follicular antrum.
- The **cumulus oophorus** is the additional cluster of cells that further surround the oocyte and anchor the corona radiata to the zona granulosa, existing as one or multiple broad links between these groups of cells.

We can again see the theca surrounding the follicle. They serve the same functions but note that the next step will be ovulation. This is assisted by contraction of the **theca externa**'s smooth muscle cells, specifically under the stimulation of a surge in **luteinizing hormone** (**LH**).

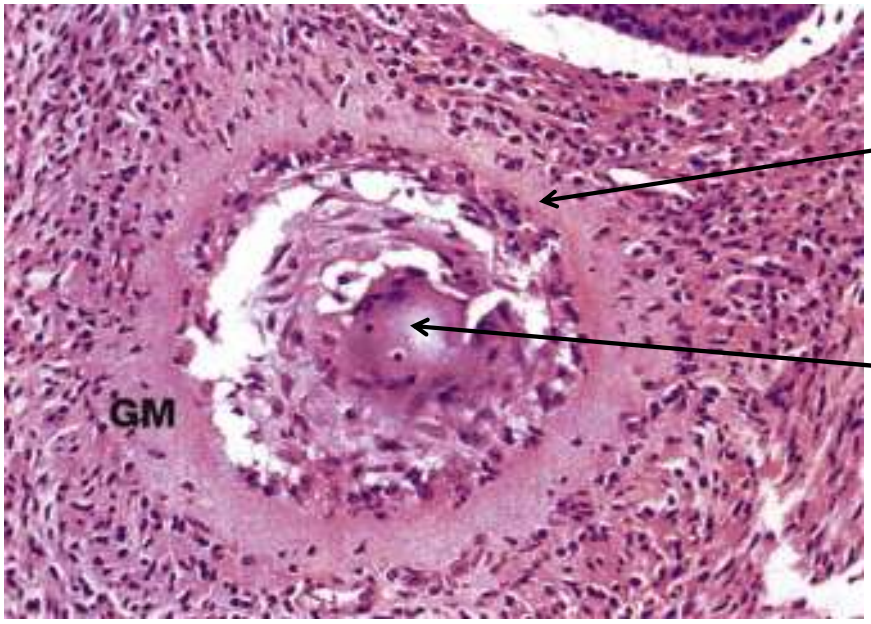
OVARIAN CORTEX: EARLY ATRETIC FOLLICLE



After the mature (Graafian) follicle stage, there are two options: either the follicle can be the "chosen one" (or occasionally two) and continue through ovulation. Or it can be one of the other ~19–29 follicles that degenerates. Follicles regularly degenerate at all stages of development, starting in fetal life and persisting throughout life. This process is termed **atresia**, and we can see the withering away of the granulosa cells and oocyte above. In smaller follicles, the oocyte degenerates and the follicular or granulosa cells (depending on the stage of development) follow suit and undergo apoptosis. These atretic follicles are usually resorbed by the stromal tissue.

In larger follicles, the oocyte degenerates independently of the granulosa cells. The oocyte takes on a smudgy appearance. The granulosa cells become discohesive with pyknotic nuclei. Macrophages infiltrate the degenerating follicle, particularly to resorb the zona pellucida. The basement membrane of the granulosa cells may increase in thickness and remain as a glassy membrane, shown on the next slide.

OVARIAN CORTEX: LATE ATRETIC FOLLICLE

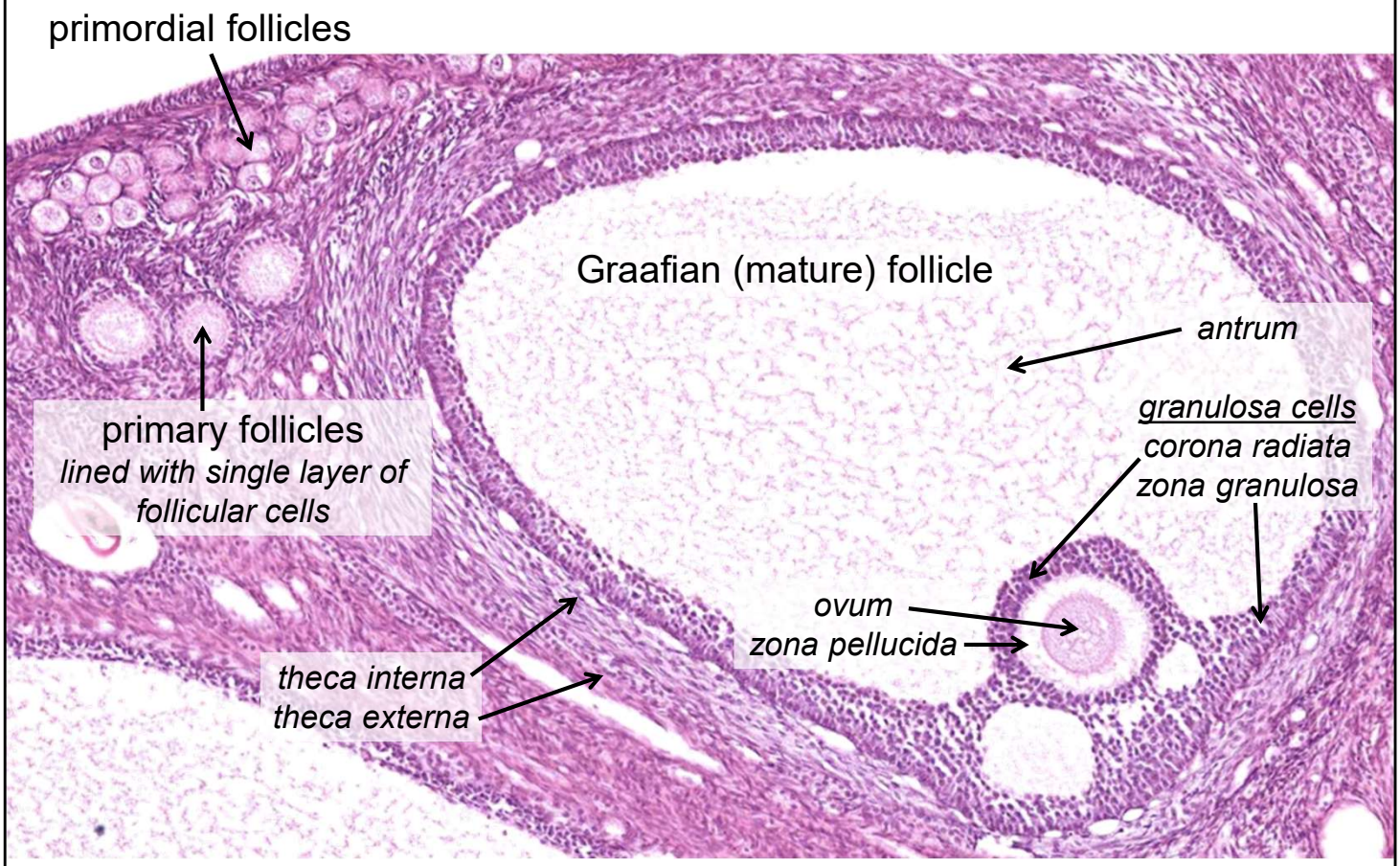


glassy membrane
thickened basement membrane formed between granulosa cells and theca interna

oocyte
degenerated

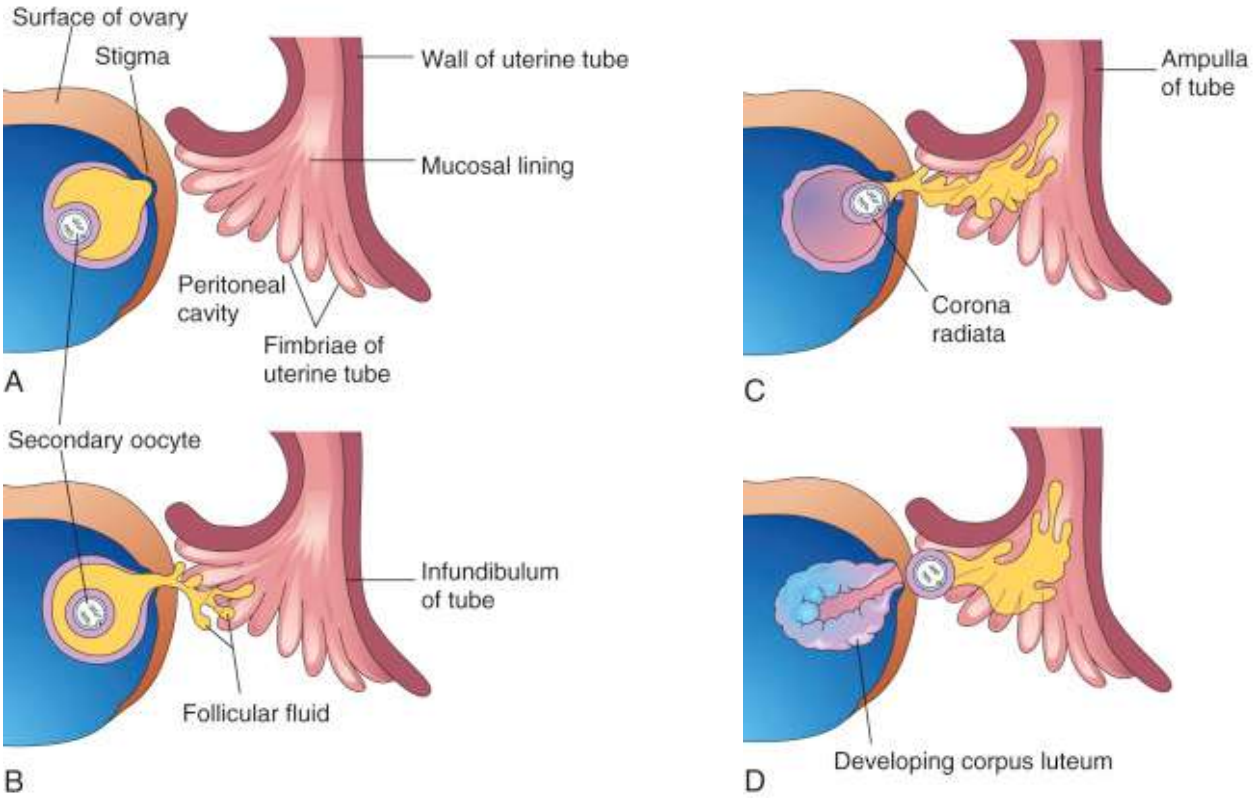
And as it nears the end of atresia, we can see that the granulosa cells and oocyte have essentially disappeared. The former follicle is surrounded by a thickened layer of basement membrane termed a **glassy membrane** between the degenerated granulosa cells and the theca.

OVARIAN CORTEX: SUMMARY OF FOLLICLES



This slide just presents the follicles in various stages of development, all on the same slide. Please refer to the previous few slides for a more detailed discussion of each stage.

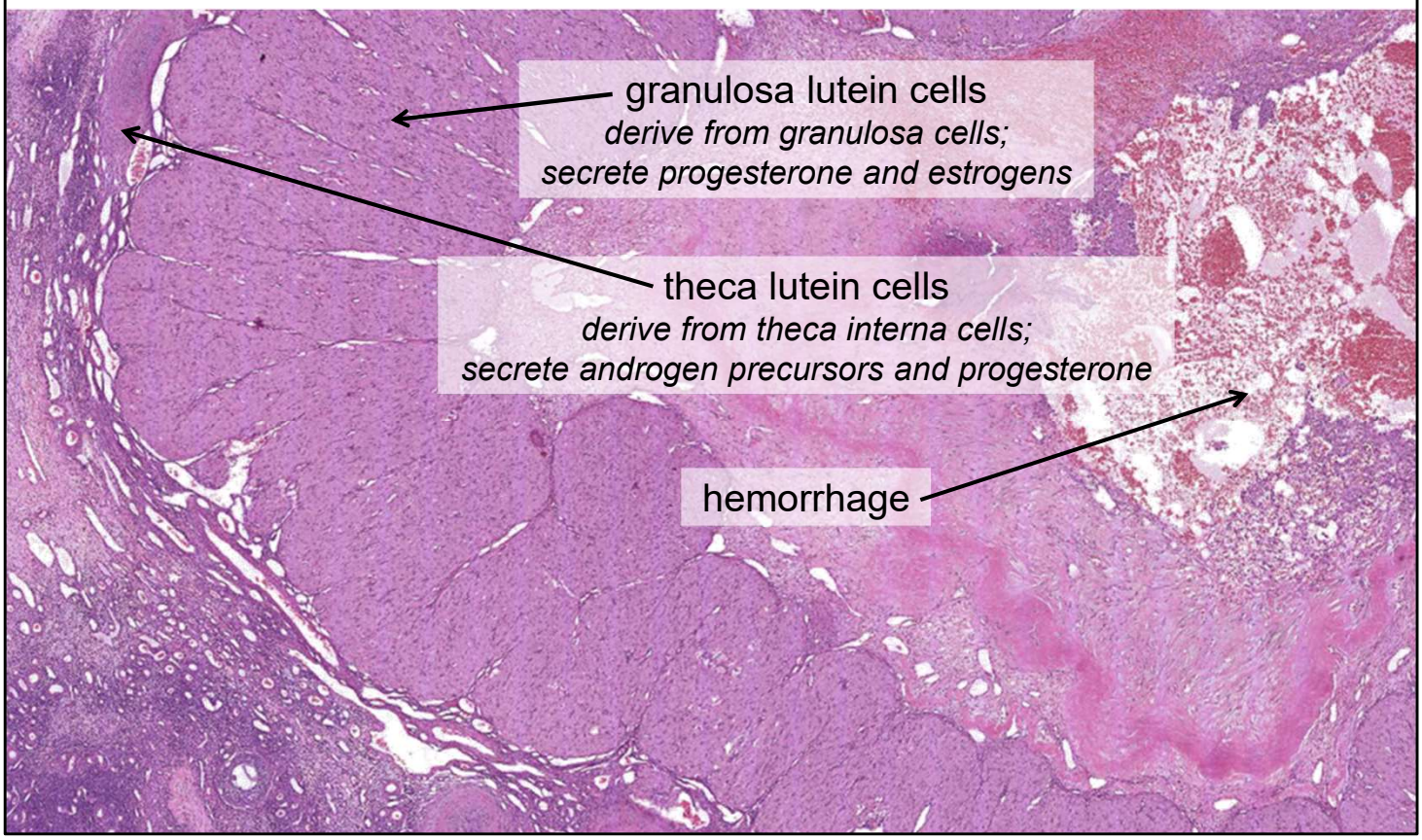
OVULATION



At day 14 (assuming a 28-day cycle), a surge of **luteinizing hormone** (**LH**) will stimulate **ovulation**, the release of the oocyte from the ovary. The mature follicle, now containing an oocyte arrested in **metaphase II**, will breach the surface layers of the tunica albuginea and the mesovarium, resulting in a **stigma** (**macula pellucida**) that appears as a transparent swelling in the ovarian surface (see image from title slide). It will rupture, releasing the follicular fluid. The oocyte and its associated cumulus oophorus (note: not just the corona radiata, as the image would have you believe) will disconnect from the zona granulosa cells and will encounter the **fimbriae** of the **oviduct** (**uterine** or **Fallopian tube**). The epithelium lining the oviduct is ciliated, which helps guide the follicular fluid and oocyte into the **infundibulum** of the oviduct—more on this in the next section.

So, what's left behind? Some granulosa cells and the surrounding theca. What are those tissues to do?

OVARIAN CORTEX: CORPUS LUTEUM

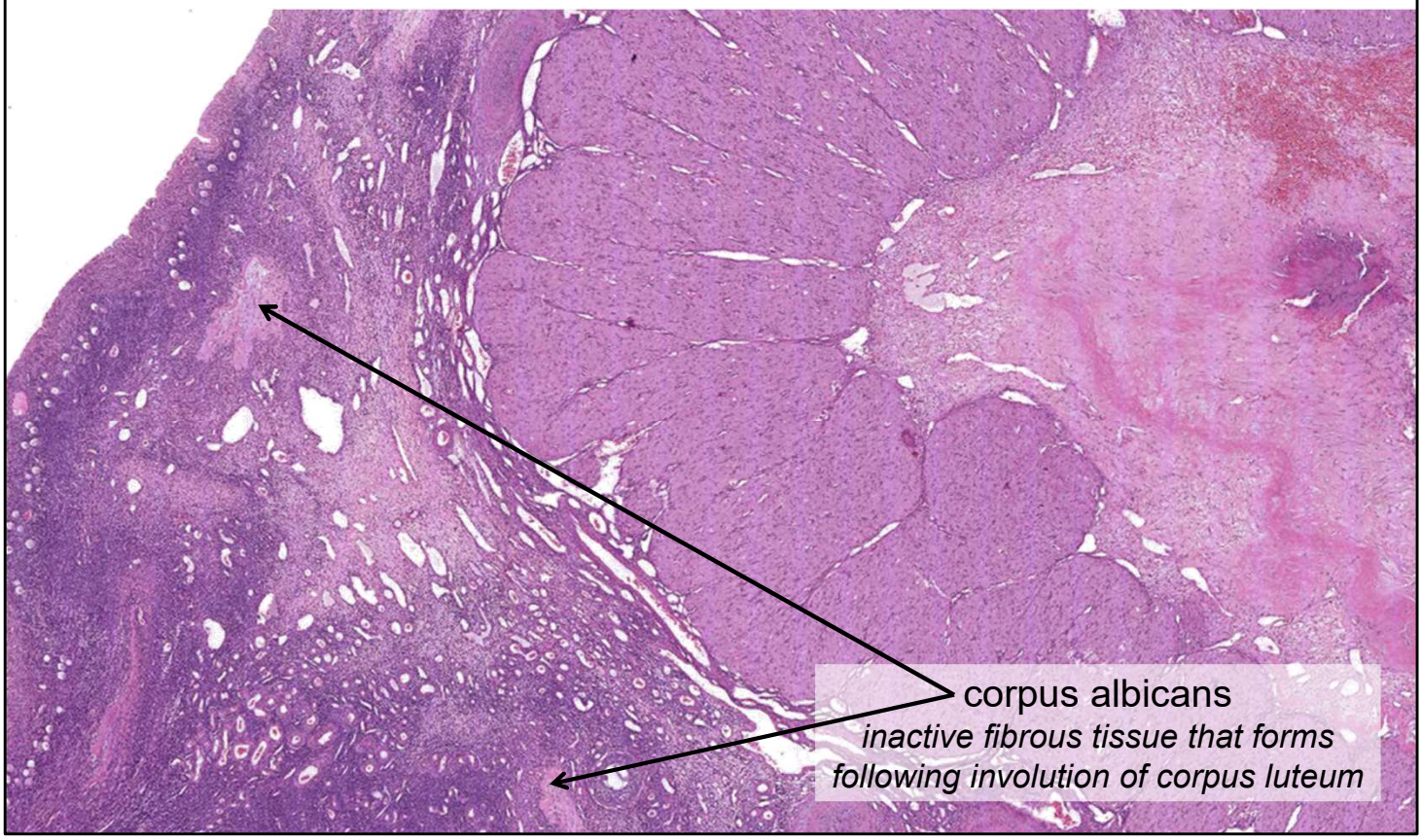


Well, they turn into the **corpus luteum** ("yellow body" in Latin). The granulosa cells mature into **granulosa lutein** cells (basically, "yellow granulosa cells"). They become more active as steroid hormone-producing cells, producing **progesterone** and **estrogens** in increased concentration (as well as **inhibin A**). We'll examine the roles of these hormones on the endometrial lining, as well as the menstrual cycle overall, in a later section. The **theca lutein** cells derive from the prior theca interna and secretes androgen precursors that are converted by the granulosa lutein cells into progesterone (predominantly) and estrogens; they also produce progesterone themselves.

We also see the site of the former oocyte and cumulus oophorous, now marked with **hemorrhage**.

If fertilization occurs, the corpus luteum will be maintained. This is accomplished by **human chorionic gonadotropin** (***β*-hCG**), which is a luteinizing hormone analogue (that is, it will stimulate the same receptor, keeping the corpus luteum alive). *β*-hCG is produced starting at the 8-cell stage of development and will be secreted into the bloodstream upon implantation of the subsequent blastocyst.

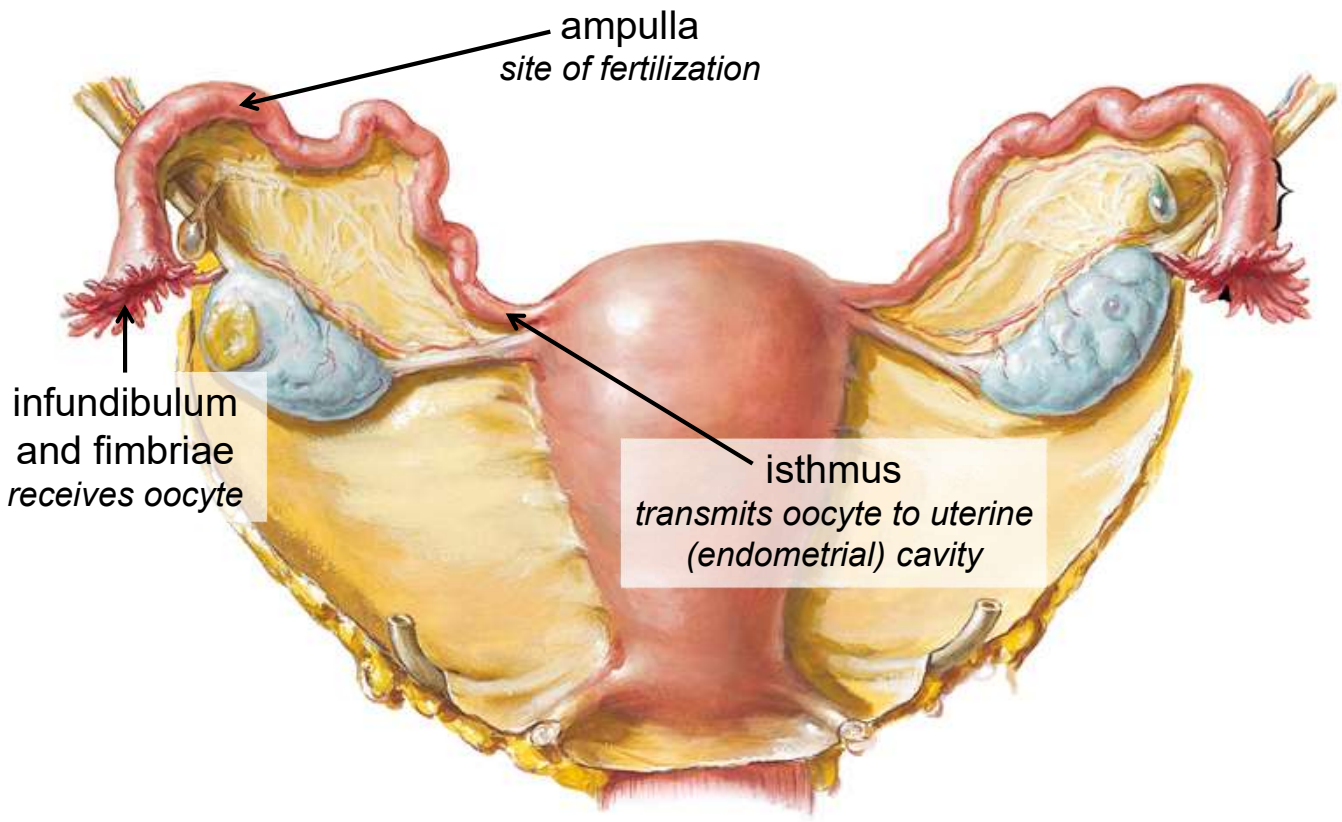
OVARIAN CORTEX: CORPUS ALBICANS



If fertilization does *not* occur, the corpus luteum will begin to involute on its own after about 10 days. In so doing, it becomes a small, scarred structure termed the **corpus albicans** ("white body" in Latin). There's functionally a "time bomb" in the corpus luteum, and the only override signal is *β*-hCG.

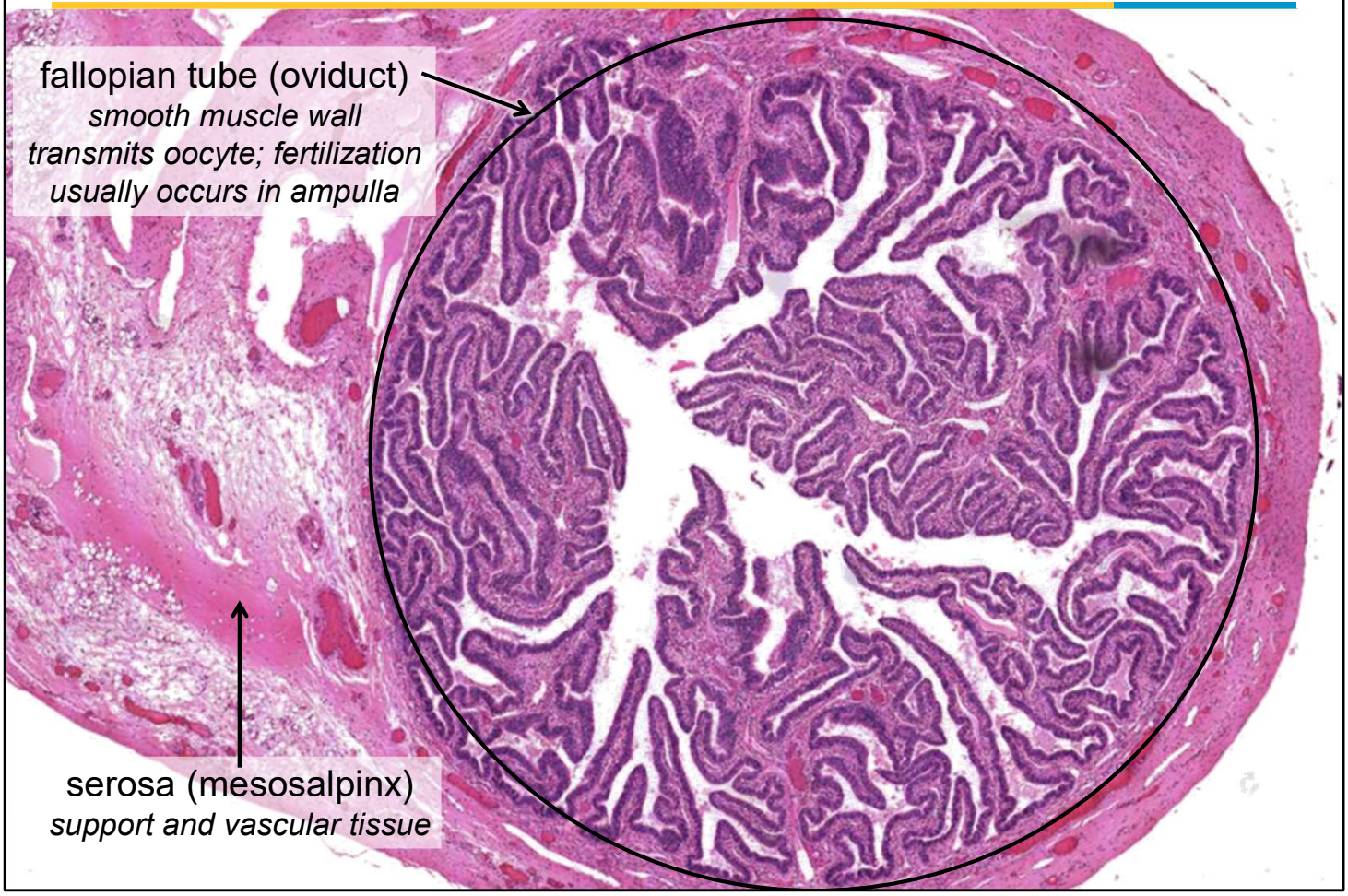
The Uterine Tube (Fallopian Tube or Oviduct)

PARTS OF THE UTERINE (FALLOPIAN) TUBE (OR OVIDUCT)



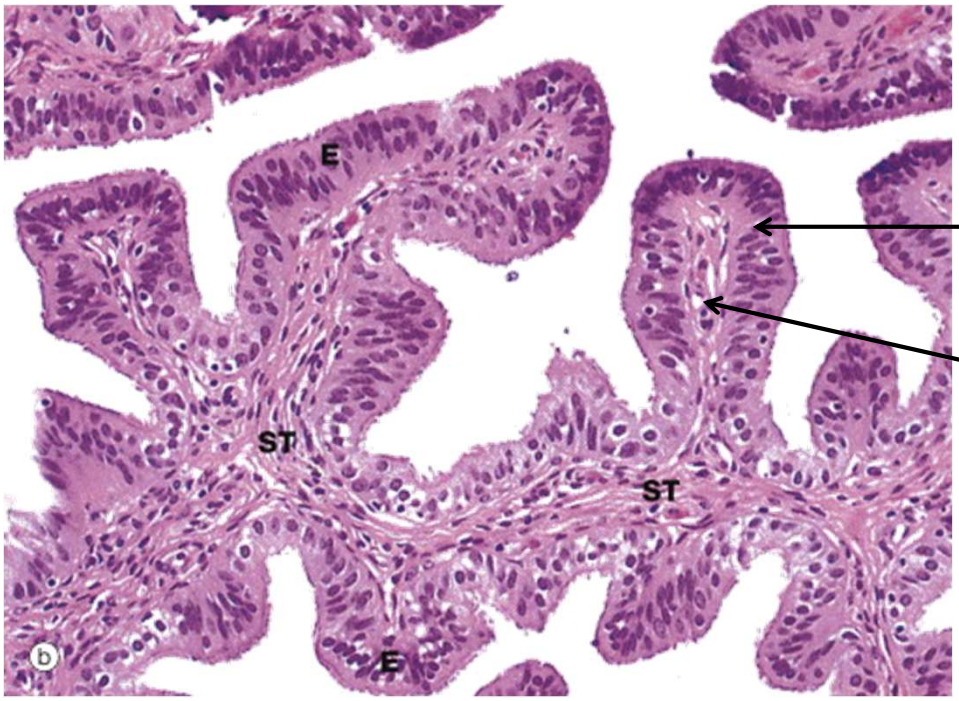
We move on as the oocyte has been ovulated to travel down this structure of many potential names: the **uterine tube**, **Fallopian tube**, or **oviduct**. It begins with the **fimbriated** end of the **infundibulum**, as described before. From there, it tapers down a long **ampulla**, which is also the site where fertilization generally occurs. At it approaches the uterus, it tapers further into a narrow **isthmus**, which meets the uterus at the **uterotubal junction**. There, it extends through an **intramural segment** running through the smooth muscle wall of the uterus (not shown) to reach the endometrial cavity. In total, the oviduct is approximately 12 cm in length.

UTERINE TUBE (OVIDUCT): STRUCTURE



The oviduct has a complex, branched mucosa appreciable at low power. (One wonders how the ovum successfully navigates this complex labyrinth so consistently!) It is suspended by connective tissue in a **serosa**, which—like the ovary—is lined with a portion of peritoneum within the broad ligament. Deep to the serosa is smooth muscle, organized into inner circular and outer longitudinal layers, reminiscent of the gastrointestinal tract. This fold of peritoneum supporting the tube is called the **mesosalpinx**. It will also be lined with a simple squamous to low cuboidal epithelium (not well seen here).

UTERINE TUBE (OVIDUCT): MUCOSA



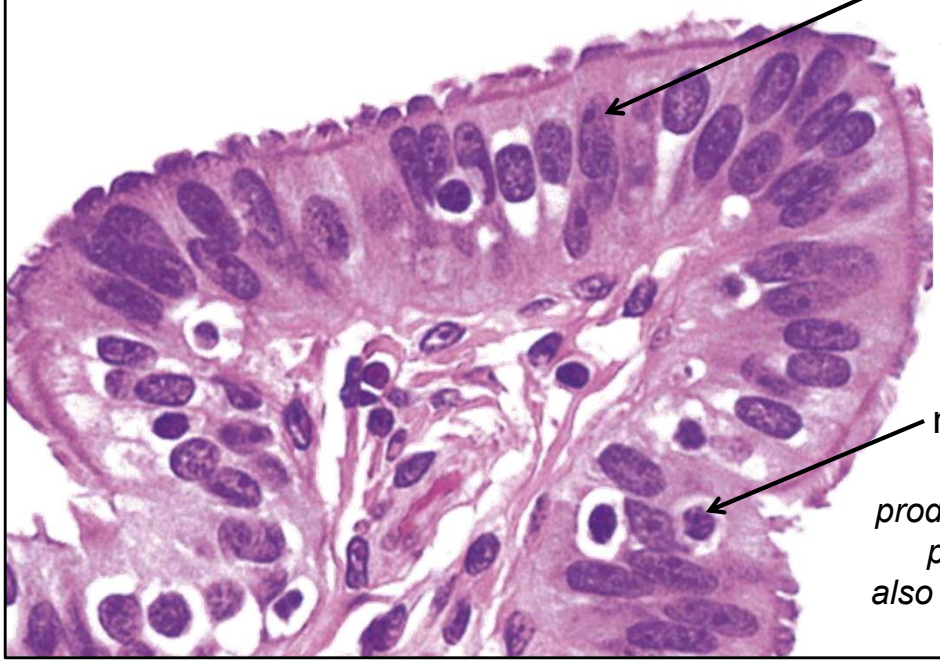
epithelium
simple columnar

lamina propria
contains fibrovascular cores

Zooming in on the structure of this branched mucosa, we see that it consists of a simple columnar epithelium overlying a **lamina propria** containing loose connective tissue. That loose connective tissue contain fibrovascular cores (supportive proteins, like collagen, and blood vessels). The epithelium may not look ideally simple columnar—in this shot, it looks downright pseudostratified or stratified columnar even. But rest assured, every one of those cells is in contact with the basement membrane and extends to the surface. That's a simple columnar epithelium.

The mucosa of the oviduct is hormonally responsive, hypertrophying during the follicular phase and atrophying during the luteal phase.

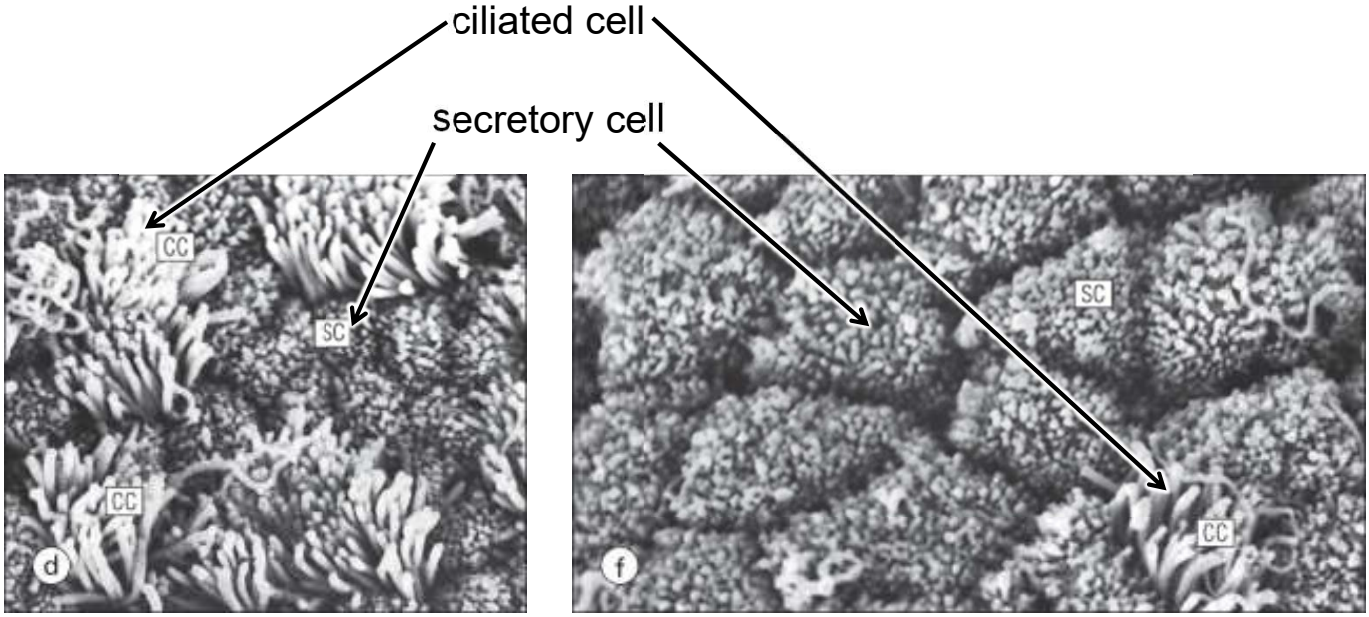
UTERINE TUBE (OVIDUCT): EPITHELIUM



There are two major cell types that line the oviduct. First are **ciliated cells** that sweep the fluid within the uterine tube toward the endometrial (uterine) cavity. The other are **non-ciliated columnar** (**peg**) **cells**, which produce substances that support the oocyte and nourish and facilitate activation of spermatozoa. As we will see on the next slide, the distribution of these cells changes over the course of the tube.

During the menstrual cycle, estrogens increase ciliogenesis (proliferation of the ciliated cells), whereas progesterone increases secretory cell number.

CELL TYPES OF THE UTERINE TUBE (OVIDUCT)



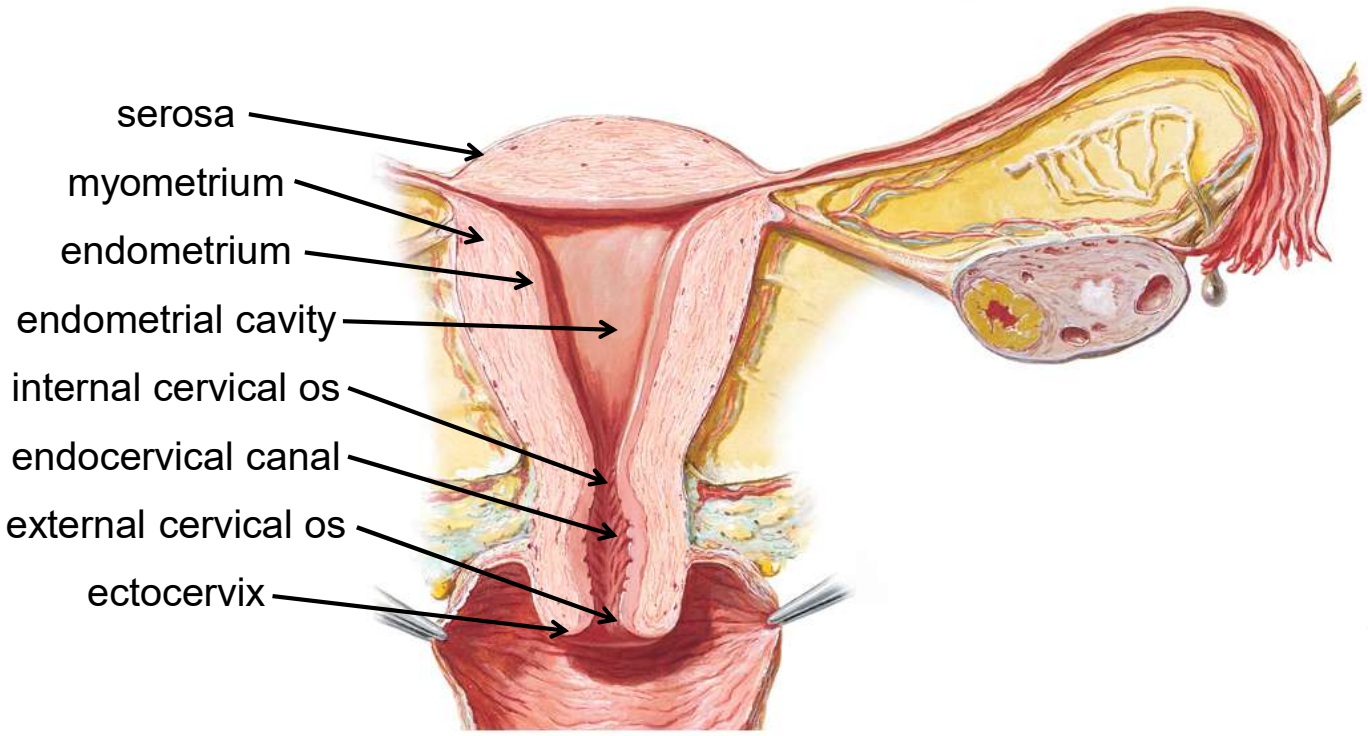
infundibulum, ampulla
more ciliated cells;
fewer secretory cells

isthmus
more secretory cells;
fewer ciliated cells

We see both cell types down the length of the uterine tube, but their proportion changes. Toward the ovarian end (infundibulum [including fimbriae] and ampulla), there are more ciliated cells, generating a current to draw in the oocyte and direct its motion toward the endometrial cavity. There are fewer secretory cells. The ratio slowly shifts along the ampulla, so that by the time one reaches the isthmus, there are more secretory cells and fewer ciliated ones.

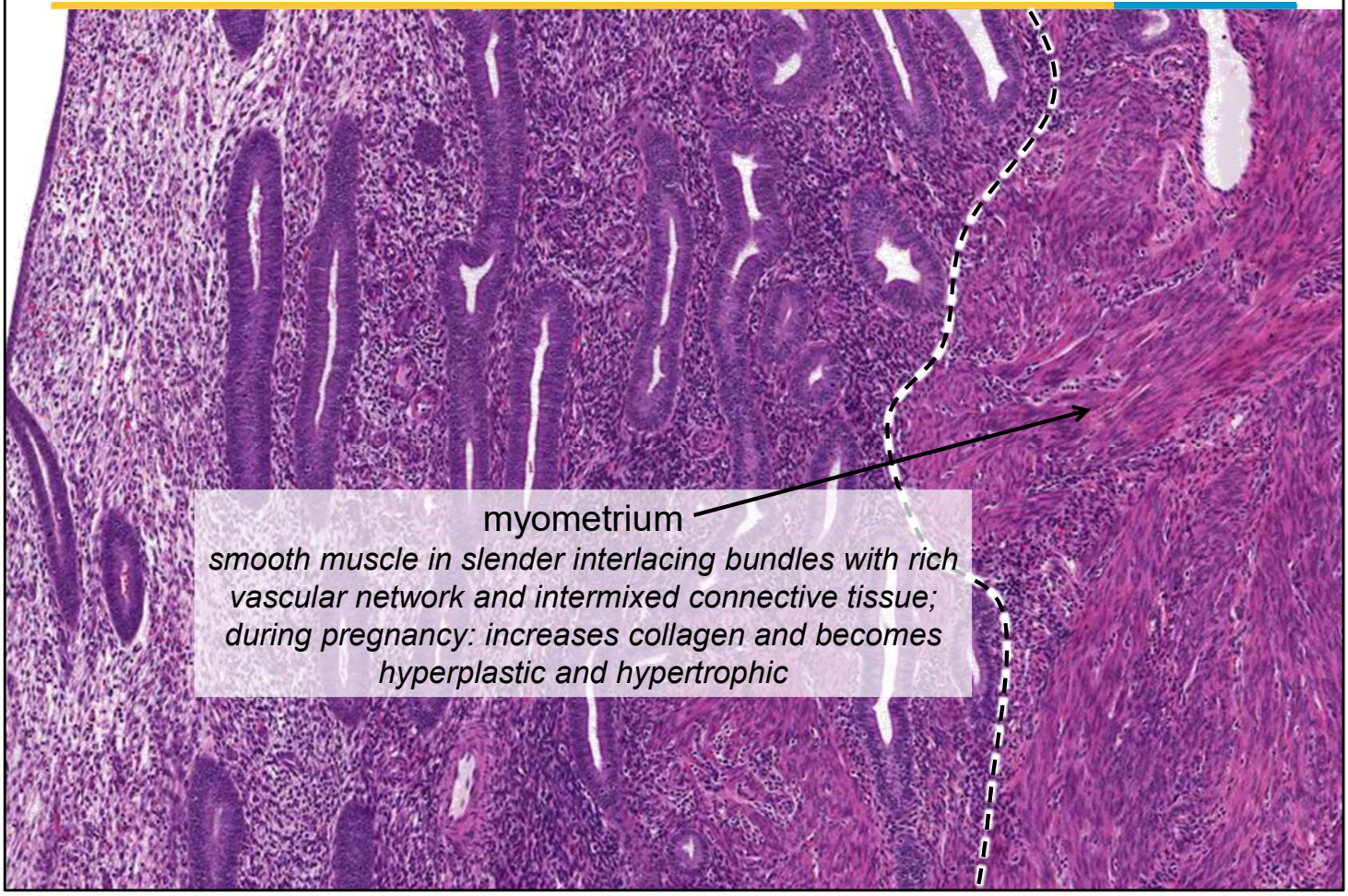
The Uterus

PARTS OF THE UTERUS



As the ovum exits the oviduct, it enters the **endometrial** (**uterine**) **cavity**, the deepest portion of the organ. The **uterus** can be divided into three tissue layers. Externalmost is the **serosa**, which corresponds to the **mesometrium**—the mesentery supporting the organ as part of the broad ligament (itself, in turn, just considered a portion of the peritoneum in this region). Most of the uterine wall is composed of the **myometrium**, which consists of woven fascicles of smooth muscle. Deepest, again, is the endometrial (uterine) cavity, lined with the glandular **endometrium**. The endometrial cavity is tapered in the direction of the **cervix**. In cross-section, we see the transition point from the endometrial cavity to the **endocervical canal**, called the **internal** (**cervical**) **os**. The distal end of the endocervical canal empties into the proximal vagina through the **external** (**cervical**) **os**. On the external surface of the cervix is the **ectocervix**, in direct continuity with the wall of the vagina. We will see in the next section the change in epithelial lining from the endocervix to the ectocervix—an area termed the transformation zone, which will be of great clinical importance.

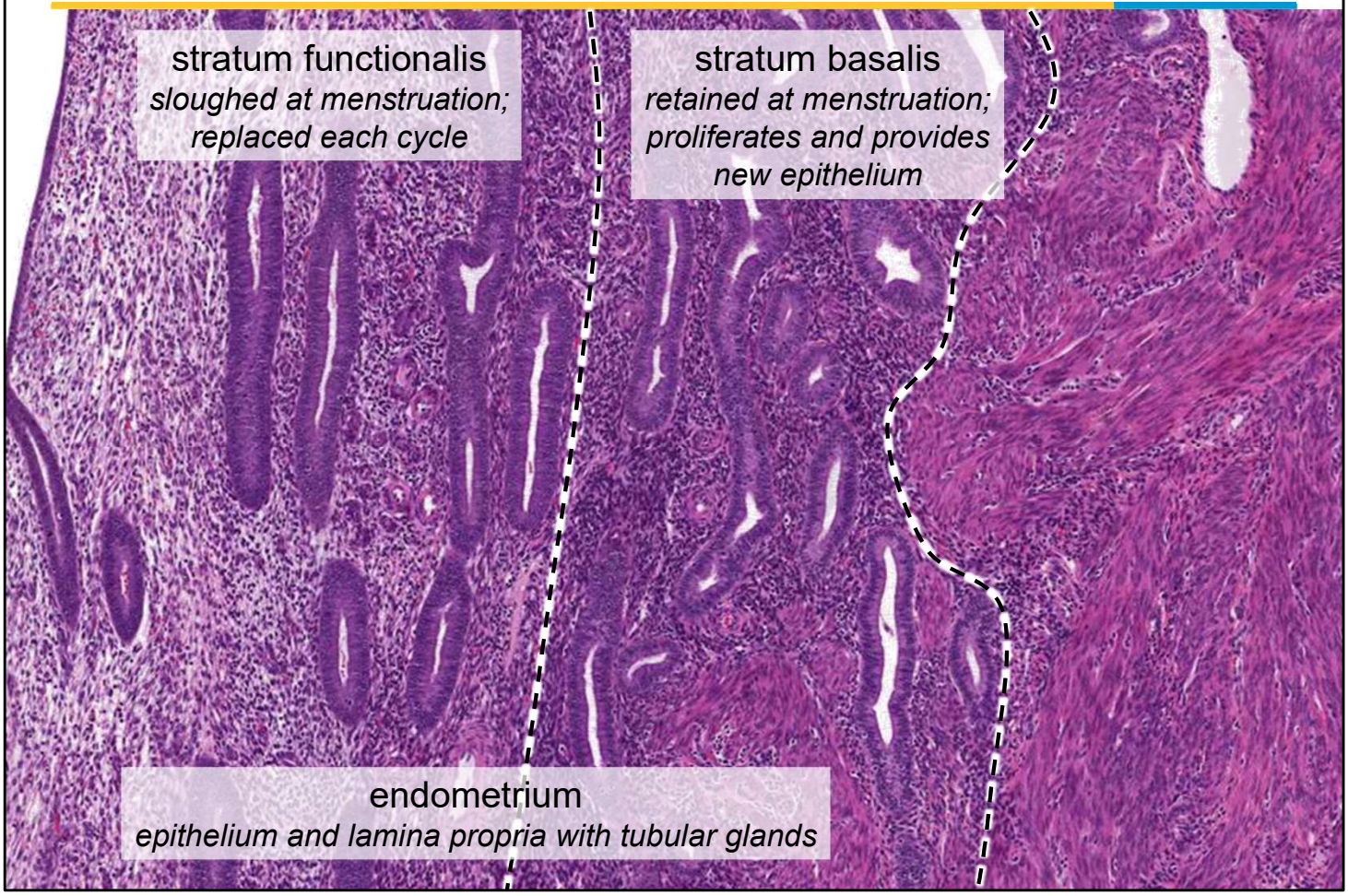
LAYERS OF THE UTERINE WALL: MYOMETRIUM



At a histological level, we start with looking at the **myometrium**; the serosa, like other mesothelia, consists of a simple squamous to low cuboidal layer lining the external surface of the organ. At a histological level, we can appreciate the interwoven or interlacing architecture of the smooth muscle fascicles, creating a somewhat “busy” or “streaming” appearance. There is an extensive vascular network in this muscle, as significant blood flow would need to be provided in cases of pregnancy, to support the placenta. This vasculature becomes more prominent during pregnancy, as do a few other changes: there is an increase in collagen deposition, and the smooth muscle fascicles increase in cell number (hyperplasia) and individual cell size (hypertrophy).

Although this slide is too zoomed in to appreciate the various layers of the myometrium, there are three: an inner and outer layer containing smooth muscle with fibers arranged roughly parallel to the long axis of the uterus, and a **stratum vasculare** sandwiched in between, containing spirally-oriented smooth muscle and large vascular plexes. Examine the low-power view of the virtual slides on Biolucida to see these somewhat poorly defined layers.

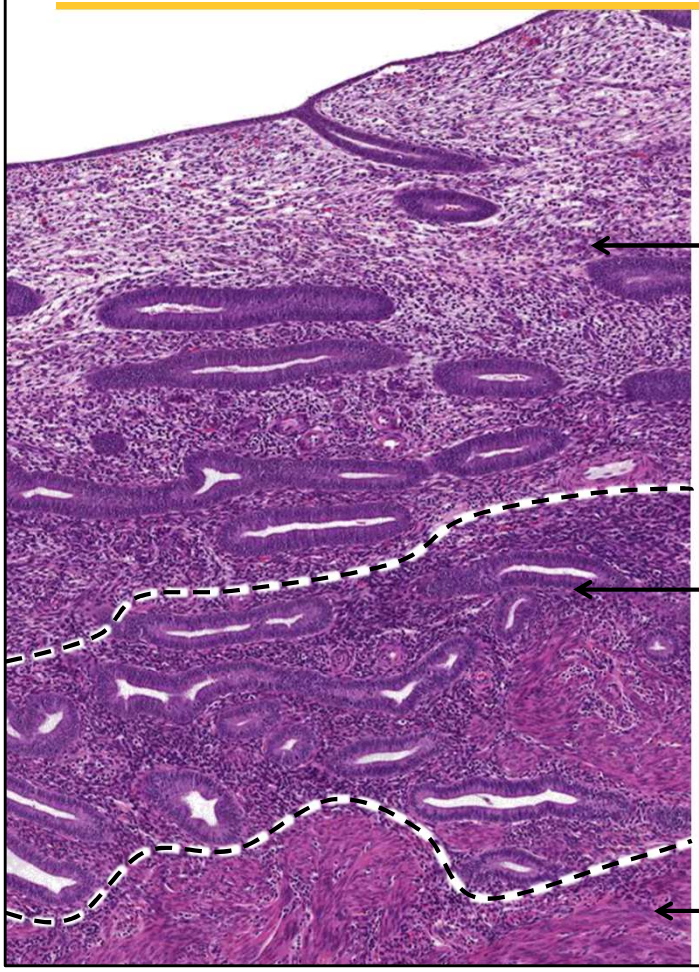
LAYERS OF THE UTERINE WALL: ENDOMETRIUM



The **endometrium** is the mucosa of the uterus, consisting of epithelium and lamina propria (loose connective tissue) with tubular to irregularly shaped glands. The specific appearance of the endometrium will change markedly during the menstrual cycle. For now, we start with an endometrium in proliferative phase (described further on the next slide) to demonstrate the two layers of the endometrium itself. The more basally oriented layer, in immediate contact with the underlying myometrium, is the **stratum basalis**. As we will see subsequently, the stratum basalis does not change significantly in its appearance during the menstrual cycle; it retains a structure of round to tubular simple columnar glands with elongated, “cigar-shaped” nuclei. This layer is not shed during menstruation but will be able to proliferate the repopulate its overlying layer during the next menstrual cycle.

The more superficial layer, which will drastically change in appearance, is the **stratum functionalis**. At the time point shown (proliferative phase), it has a similar appearance to the stratum basalis. But this layer will develop during the menstrual cycle and ultimately be shed if fertilization and implantation do not occur, representing the tissue sloughed during menstruation.

LAYERS OF THE ENDOMETRIUM: PROLIFERATIVE PHASE



associated with estrogen secretion from developing follicles

stratum functionalis

small rings of glandular cells in cellular stroma; glands extend to surface

stratum basalis

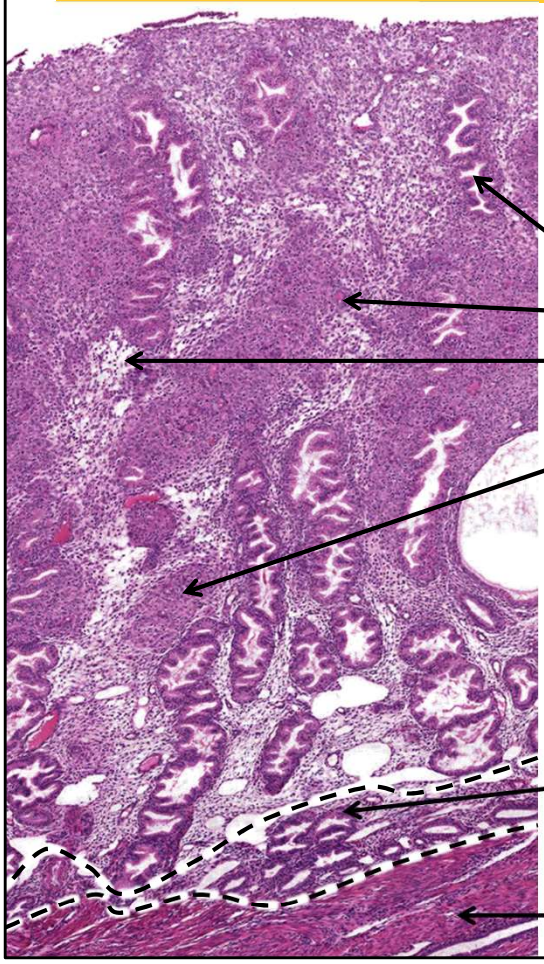
small rings of glandular cells in cellular stroma; immediately superficial to myometrium

myometrium

We'll summarize the full changes seen in both the ovary and endometrium in the menstrual cycle in two slides, but let's start with the endometrium in **proliferative phase** from approximately day 6 to 14, assuming a 28-day cycle (note that **menses** [**menstruation**] occurs from day 1 to 5 but will be considered at the *end* of this narrative in this session.) During proliferative phase, both the **stratum basalis** and **stratum functionalis** consist of small rings or tubules of simple columnar epithelial cells embedded in relatively cellular stroma (the **lamina propria**). At least some of the glands in the stratum functionalis can be seen extending to the surface. These glands are actively proliferating, especially in the stratum functionalis; this is due to estrogen-mediated hyperplasia. The **estrogen** itself is synthesized by the ovarian granulosa cells, as described previously.

The arteries of the myometrium give rise to **radial arteries** that penetrate the endometrium. These radial arteries give off **spiral arteries** that supply an extensive capillary field that contains occasional enlarged regions of blood called lacuna. These spiral arteries will become much more apparent during the secretory phase of the menstrual cycle, as shown on the next slide.

LAYERS OF THE ENDOMETRIUM: SECRETORY PHASE



associated with progesterone secretion from corpus luteum after ovulation

stratum functionalis

complex glands and glycogen-rich secretory mucus

coiling of spiral arteries

intercellular edema

pink stroma (predecidual change)

stratum basalis

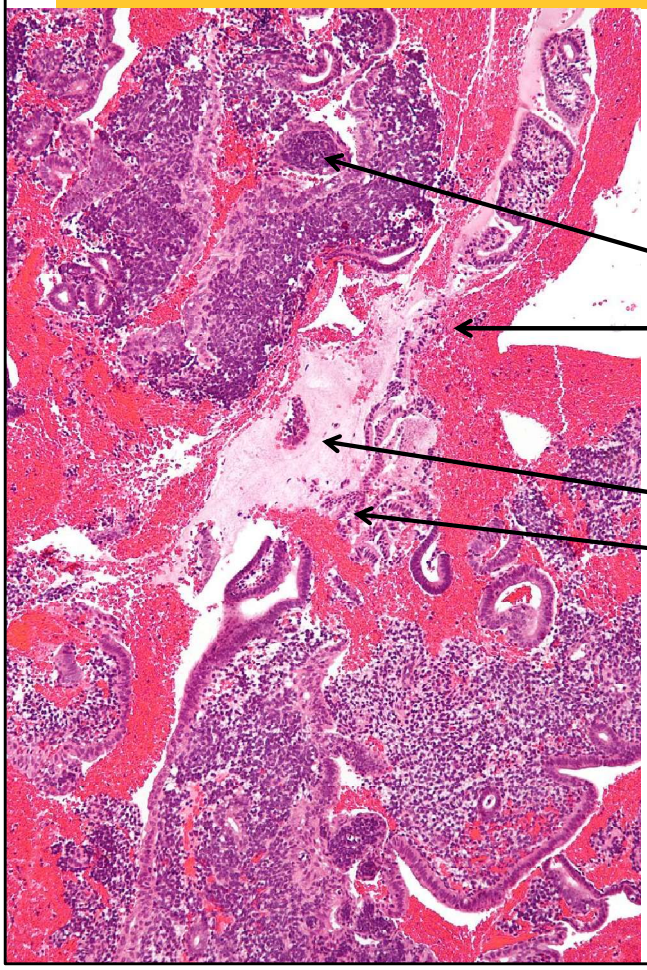
small rings of glandular cells in cellular stroma; immediately superficial to myometrium

myometrium

During the **secretory phase** (days 14 through 28), the stratum functionalis alters its appearance to form complex, corkscrew-shaped glands with much paler cytoplasm. This change in cytoplasmic color reflects the synthesis of glycogen-rich mucus. The arteries feeding the stratum functionalis, **spiral arteries**, become progressively coiled and can be seen as many adjacent cross-sections in histological preparations. They also elongate to reach the surface of the endometrium. Later in secretory phase, some areas of the stroma begin to demonstrate intercellular edema—the accumulation of fluid between the stromal cells, giving it a decreased density and increased white space between cells. Finally, in some areas of the stroma, **predecidual change** is seen (the **decidua** is the endometrial contribution to the placenta if fertilization and implantation occur), which manifests as an increase in eosinophilic cytoplasm and prominent cell borders.

Note that the changes described here occur as a trend across secretory phase; many outside resources will subdivide secretory phase into early, middle, and late portions based on the prominence and progression of these changes. The primary hormonal trigger for these changes is the increased concentration of **progesterone** secreted by the corpus luteum—the granulosa and theca cells left behind in the ovary after ovulation. Estrogen also plays a role in further proliferation.

LAYERS OF THE ENDOMETRIUM: MENSES



withdrawal of progesterone leads to deterioration and sloughing of the stratum functionalis

stratum functionalis

endometrial stromal condensation

hemorrhage

neutrophilic infiltration (not seen in this specimen)

additional findings in this curettage

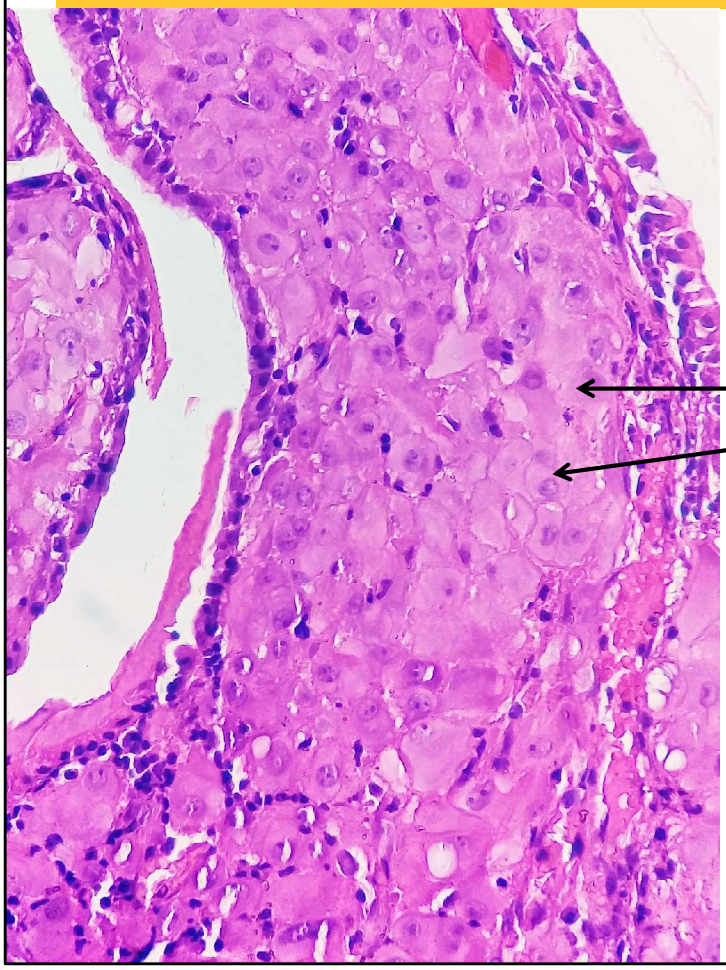
endocervical mucus

endocervical glandular cells

Finally, as the menstrual cycle restarts, **menses** (**menstruation**) occurs, roughly between day 1 and 5 of the cycle. If the corpus luteum is not maintained (which would require the production of β-hCG by the blastocyst/embryo, as described before), it will naturally break down and progesterone levels will drop. This withdrawal of progesterone leads to deterioration of the **stratum functionalis** and its subsequent sloughing. The stroma collapses in on itself, forming dense balls of nuclei in the process of **endometrial stromal condensation**. Neutrophilic infiltration into the tissue can be seen, although is not prominent in this specimen. Hemorrhage is also seen, as the spiral arteries break down and bleed.

Note that the example shown is from an endometrial curettage (scraping of the endometrial lining with a small surgical instrument called a curette), as opposed to a full section of the endometrial and myometrial wall. Accordingly, the stratum basalis is not appreciated in this image. We know from before that the stratum basalis would continue with its usual appearance of round to tubular glands embedded in cellular stroma. And we get to see a few additional tissues, given that we pass through the endocervix; more specifically, endocervical mucus (pale pink and wispy) and endocervical glandular cells (columnar; more pale eosinophilic cytoplasm than endometrial glands).

LAYERS OF THE ENDOMETRIUM: DECIDUA



ongoing progesterone production by corpus luteum (if stabilized by *β*-hCG) leads to decidual reaction (change)

stroma of stratum functionalis

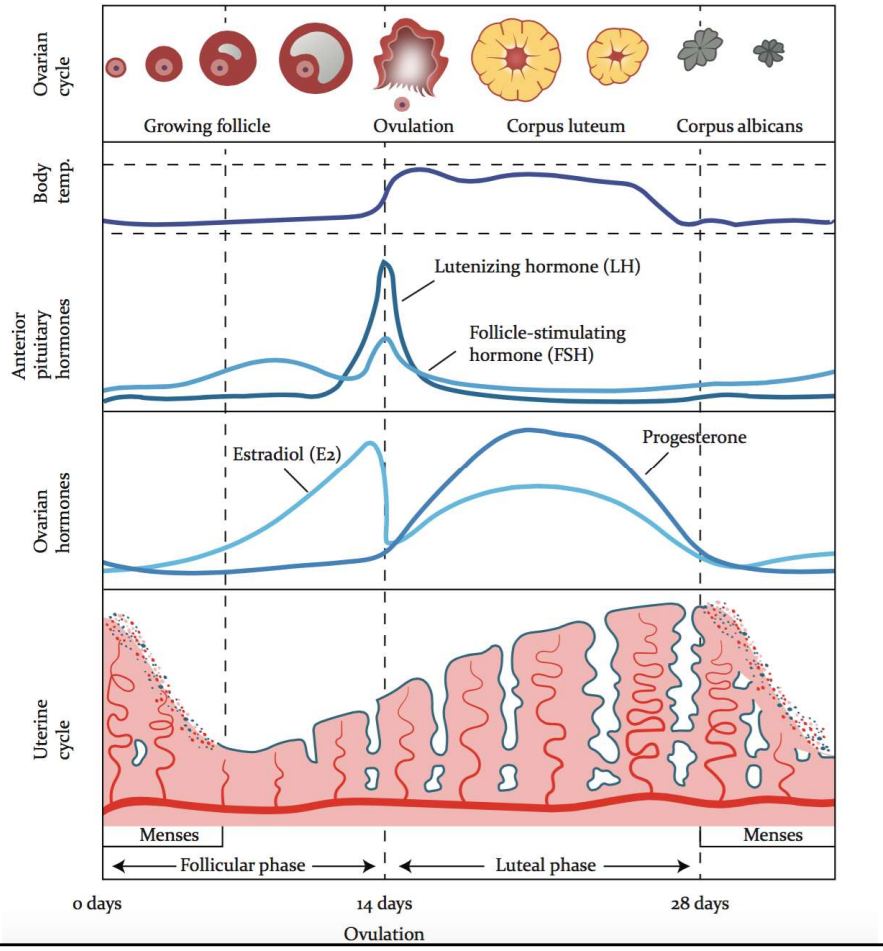
eosinophilic cytoplasm

prominent cell borders

(glycogen storage and hormone production)

In contrast, if a zygote is formed and results in an implanting blastocyst, *β*-hCG will be synthesized by the developing embryo, which will stabilize the corpus luteum. Progesterone production will increase, and predecidual change seen in secretory endometrium will give way to frank **decidua** (the endometrial portion contributing to the placenta). Notable changes follow from the description of predecidual change given before: the cells will expand in size with glassy, eosinophilic cytoplasm. Cell borders are prominent. The cells of the decidua themselves are storing glycogen and producing a wide range of hormones to facilitate implantation and support growth of the embryo.

SUMMARY: THE MENSTRUAL CYCLE



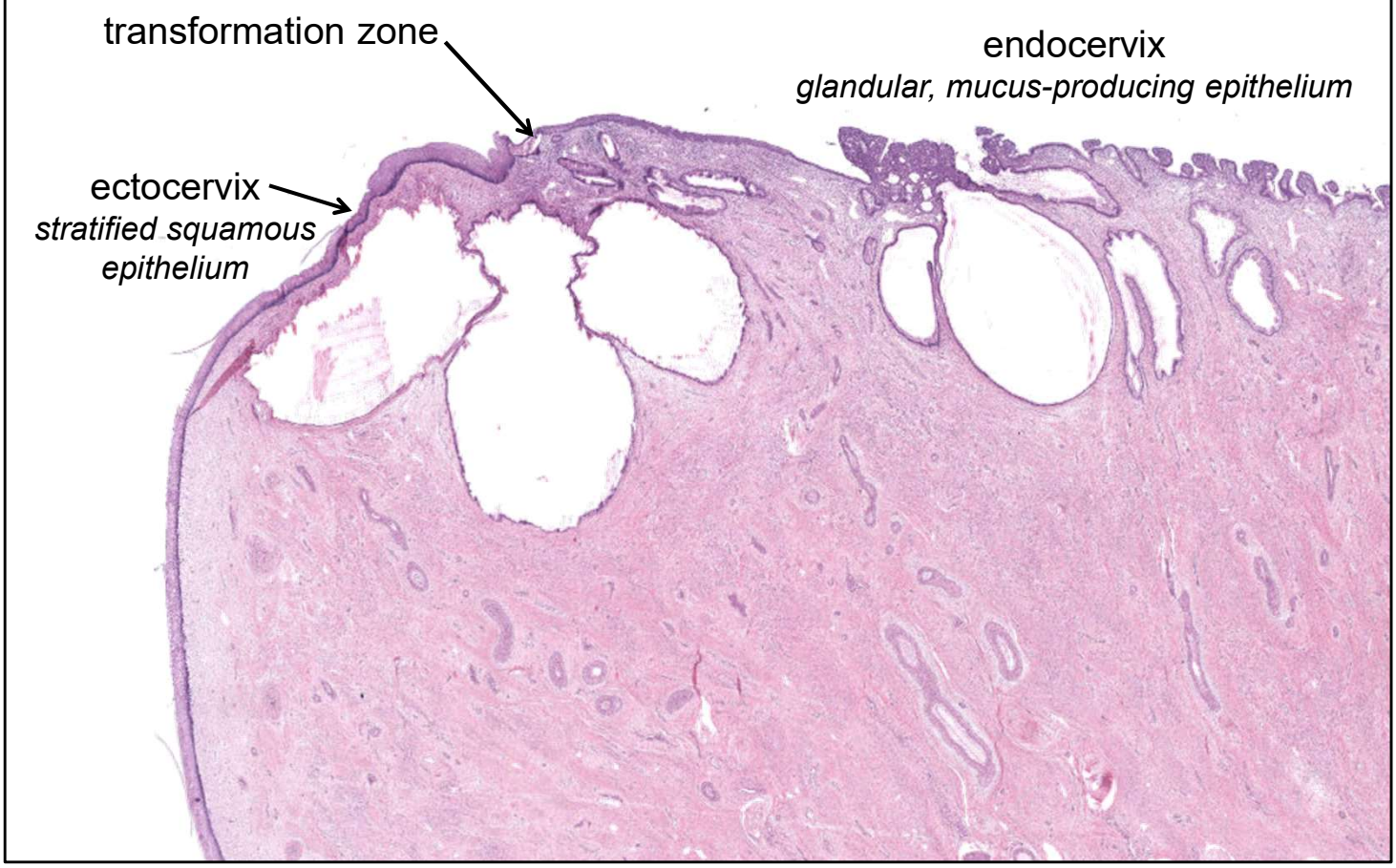
Let's summarize the changes in both the ovary and the endometrium during the menstrual cycle, to bring all the pieces together. We start at the beginning of follicular/proliferative phase, as soon as menses has completed. In **follicular phase** of the ovary, corresponding with the **proliferative phase** of the endometrium, a group of **primordial follicles** is recruited in the ovary to develop into **primary follicle**, and eventually **secondary** (**antral**) **follicles**. The beginning of the secondary (antral) follicle, where the follicular antrum can be seen, roughly corresponds to the beginning of proliferative phase in the endometrium. During this time, **follicle-stimulating hormone** encourages the proliferation of follicular, and then granulosa cells. During this part of the cycle, **luteinizing hormone** (**LH**) stimulates the theca cells to produce androstenedione, which is converted to estrogen by the granulosa cells, which express aromatase. During the early part of follicular phase, **estrogen** has a negative inhibitory effect on the release of FSH and LH from the anterior pituitary. But as a single follicle develops into a **mature** (**Graafian**) **follicle**, a threshold is reached wherein estrogen exhibits a positive feedback loop that culminates in the surge of LH that stimulates ovulation. Meanwhile, the production of estrogen has a proliferative effect on the glands of the **stratum functionalis**, leading to a gradual increase in the thickness of the endometrium.

At **ovulation**, due to the LH surge at day 14, the theca externa cells contract and facilitate the extrusion of an oocyte and its associated cumulus oophorus through the mesovarium. This oocyte will ultimately be navigated into the oviduct by the ciliated cells lining the infundibulum and its fimbriated end. Then, in the **luteal phase** of the ovary, corresponding with the **secretory phase** of the endometrium, the oocyte is transmitted down the oviduct toward the endometrial cavity. The theca cells left behind are converted into theca lutein cells; the granulosa cells become granulosa lutein cells. The structure formed of these two cell populations is the **corpus luteum**. The hormonal production shifts from estrogen production to predominantly progesterone production (although some estrogen is still produced). The progesterone facilitates the changes seen in the endometrium, including cytoplasmic changes consistent with secretion of glycogenated mucus, coiling of the spiral arterioles, stromal edema, and predecidual change. Estrogen stimulates ongoing proliferation, continuing the increase in endometrial thickness.

If the oocyte is not fertilized, there is no blastocyst/embryo to implant and produce *β*-hCG; accordingly, the corpus luteum will atrophy, progesterone levels will drop, and the stratum functionalis will be sloughed during menses as the spiral arteries close and induce ischemia in the tissue. Endometrial stromal condensation, hemorrhage, and neutrophilic infiltration may all be seen. As menses reaches its conclusion, the stratum basalis will begin the process anew in the new follicular/proliferative phase. The corpus luteum will eventually scar and be converted into a **corpus albicans**.

The Cervix and Vagina

HISTOLOGY OF THE CERVIX



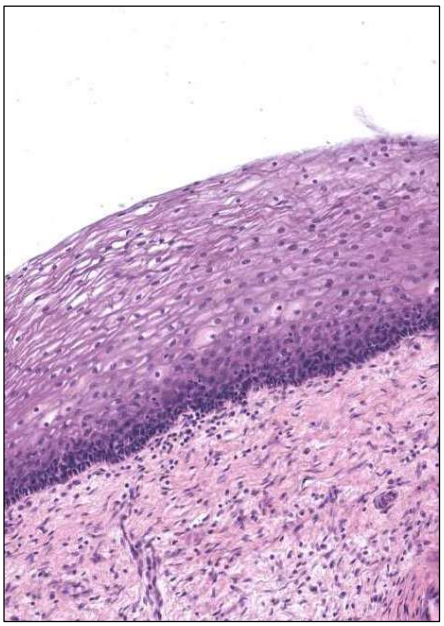
In our final section, we examine the cervix and vagina. The **uterine cervix** can be divided into an **endocervix** or **endocervical canal**, and an **ectocervix**. The **endocervix** is lined with glandular, mucus-producing cells organized into a simple columnar epithelium (seen better on the next slide). These glands also involve underlying muscle, often forming large, almost cyst-like spaces (large white spaces seen above). Production of this mucus is hormonally regulated, with more but thinner mucus being produced at mid-cycle; this facilitates sperm passage through the endocervical canal.

The **ectocervix**, in contrast, is lined with (nonkeratinizing) stratified squamous epithelium. The ectocervix is on the external surface of the cervix, in direct continuity with the vaginal canal. More technically, the ectocervix is continuous with the vaginal wall lining the fornices.

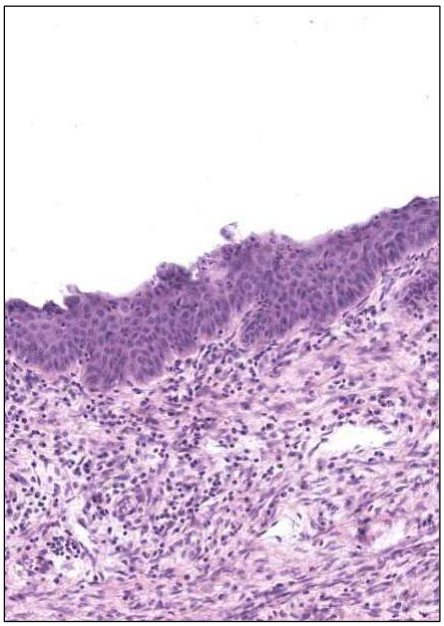
Between these two is the **transformation zone**, which contains cells that originally were glandular and mucus producing but that exhibit squamous metaplasia. This thin portion of tissue represents the dividing line between endo- and ectocervix and has great clinical importance. This area—the transformation zone—is the location where human papillomavirus (HPV), the #1 cause of cervical cancer in the United States, initiates its infection.

The underlying stroma of the uterine cervix is composed predominantly of dense, fibrous connective tissue with a minor contribution of smooth muscle. It is rich in elastic fibers, enabling cervical effacement (thinning) and dilation (opening) during labor and delivery.

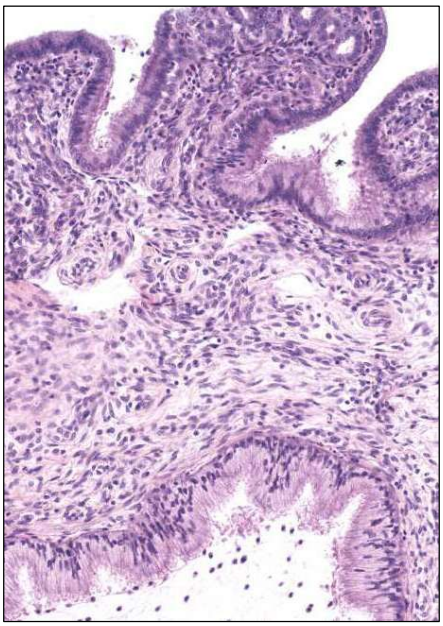
HISTOLOGY OF THE CERVIX: TRANSFORMATION ZONE



ectocervix
nonkeratinizing stratified squamous epithelium



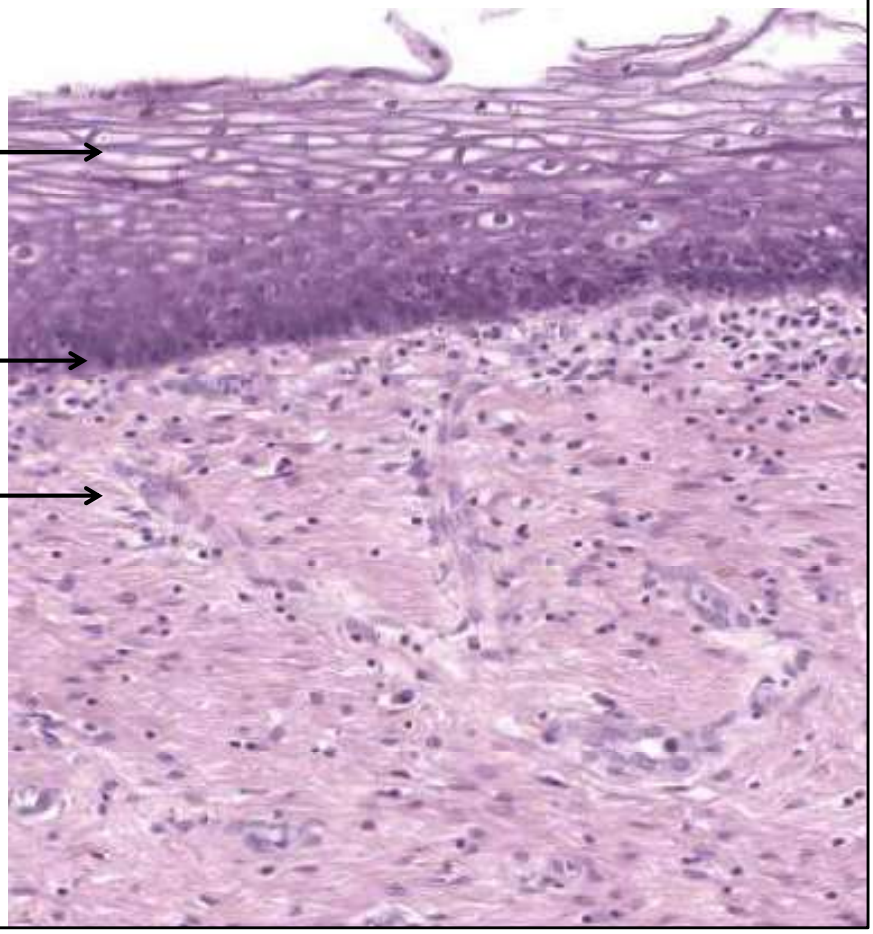
transformation zone
squamous metaplasia



endocervix
glandular, mucus-producing epithelium

Here, we've zoomed in more to appreciate the different parts of the cervix. As mentioned previously, the **ectocervix** is lined with nonkeratinizing stratified squamous epithelium. It is glycogenated, leading to the pale spaces seen within the cells and the "basketweave" appearance of the epithelium. The **endocervix**, in contrast, is lined with a simple columnar epithelium with mucus-producing cells; these cells demonstrate pale, eosinophilic cytoplasm. In between is the **transformation zone**, which contains cells that have changed their differentiation from glandular endocervix cells to squamous ectocervix-like cells. This location, again, is where HPV infection is initiated.

HISTOLOGY OF THE VAGINA



Finally, the vagina. The vaginal mucosa consists of a nonkeratinizing stratified squamous epithelium that is glycogenated. That is, the more superficial cells are packed with glycogen, which gives them empty white spaces on histology and a “basketweave” appearance. The glycogen in these cells is induced by estrogen and serves an important microbiological function: it helps support the vaginal microflora, especially *Lactobacillus* spp. The basal layer sits atop a basement membrane, which separates it from the **lamina propria**. The lamina propria consists of loose connective tissue without any appreciable glands. It contains elastic fibers and may show lymphoid nodules. Note that the source for the mucosa-lining mucus in the vagina is the (endo)cervix, which itself produces a very large volume of mucus.

LEARNING OBJECTIVES

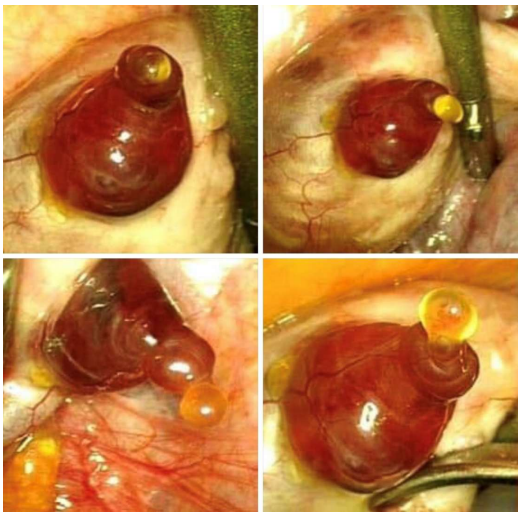
By exam time, you should be able to do the following:

- Discuss the structural and functional anatomy of the ovary and organs involved in ovum release or implantation, including how they change over different life stages and the hormonal regulation of these changes.
- Describe and identify the cells and tissues of the ovary and organs involved in ovum release or implantation by histology.
- Discuss the process of oogenesis from primordial follicle through mature oocyte, including the associated morphologic changes.
- Compare and contrast how the ovary and organs involved with ovum release or implantation function as, and are regulated by, endocrine glands; and identify the impact of hormonal change via microscopy.

Starting in academic year 2025–2026, this learning activity was changed from a histology lab to a lecture. This was done in response to student requests for more structure regarding this content. However, the resources from the session as a histology lab still exist!

The materials are included on Canvas as optional, supplementary material that give you an opportunity to practice the content within this session.

Histology of the Ovary and Organs of the Ovum Pathway



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Image: laparoscopic view of an ovary during ovulation, with the stigma visible and extruding follicular fluid.